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BEST PRACTICES IN CRAFTING

DISSERTATION



Ts Dr Noureen Talpur
Senior Lecturer
Department of Computing

DR NOUREEN TALPUR

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Ts DR NOUREEN TALPUR

Senior Lecturer | Universiti Teknologi PETRONAS (UTP)
Core Expert | Center for Resilient Data Analytics and Computational Systems (CERDAS)

EDUCATION

PhD IT, Universiti Teknologi PETRONAS, Malaysia

MSc. IT, Universiti Tun Hussein Onn Malaysia, Malaysia

B.CS, Muhammad Ali Jinnah University, Pakistan

EXPERIENCE

- Graduate Assitant - 5 years
- Teaching Experience - 6 years

RESEARCH / EXPERTISE

- Artificial Intelligence and Machine Learning for Smart Communities
- Data Science (Big Data Analytics, Data quality assessment, data mining)
- Optimization, prediction, classification, and feature selection

CONSULTANCY AND TRAINING

- Master the Basics of Big Data, AI, and Machine Learning for Tomorrow's Workforce (a fully funded program by KPT)
- Data Understanding & Data Quality Assessment
- Digital Marketing
- Problem Solving using Microsoft Excel

PROFESSIONAL BODY / MEMBERSHIP

- Professional Technologist (MBOT)
- Professional Member (IEEE)

TEACHING EXPERIENCE

1. Artificial Intelligence (using MATLAB)
2. Structured Algorithm & Programming (using Python)
3. Object Oriented Programming (using Python)
4. System Architecture
5. Big Data Analytics (using Hadoop & Map-reduce framework)
6. Discrete Mathematics
7. Statistics and Empirical Methods
8. Data Mining
9. Structured Programming
10. Embedded Systems
11. Business Intelligence

SUPERVISION

- UG students, MSc and PhD

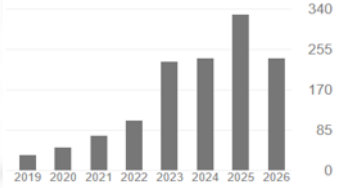


ACHIEVEMENTS

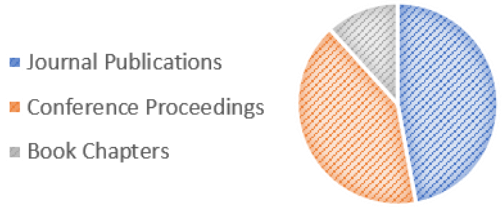
Google Scholar

Cited by

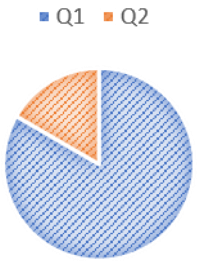
	All	Since 2021
Citations	1306	1214
h-index	12	12
i10-index	14	13



PUBLICATIONS



JOURNAL PUBLICATIONS



Research Grants Utilized = 7 | On going grants = 2

Contact | Department of Computing, UTP Malaysia ✉ noureen.talpur@utp.edu.my connect on: LinkedIn | ResearchGate | Scholar

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ICE BREAKING

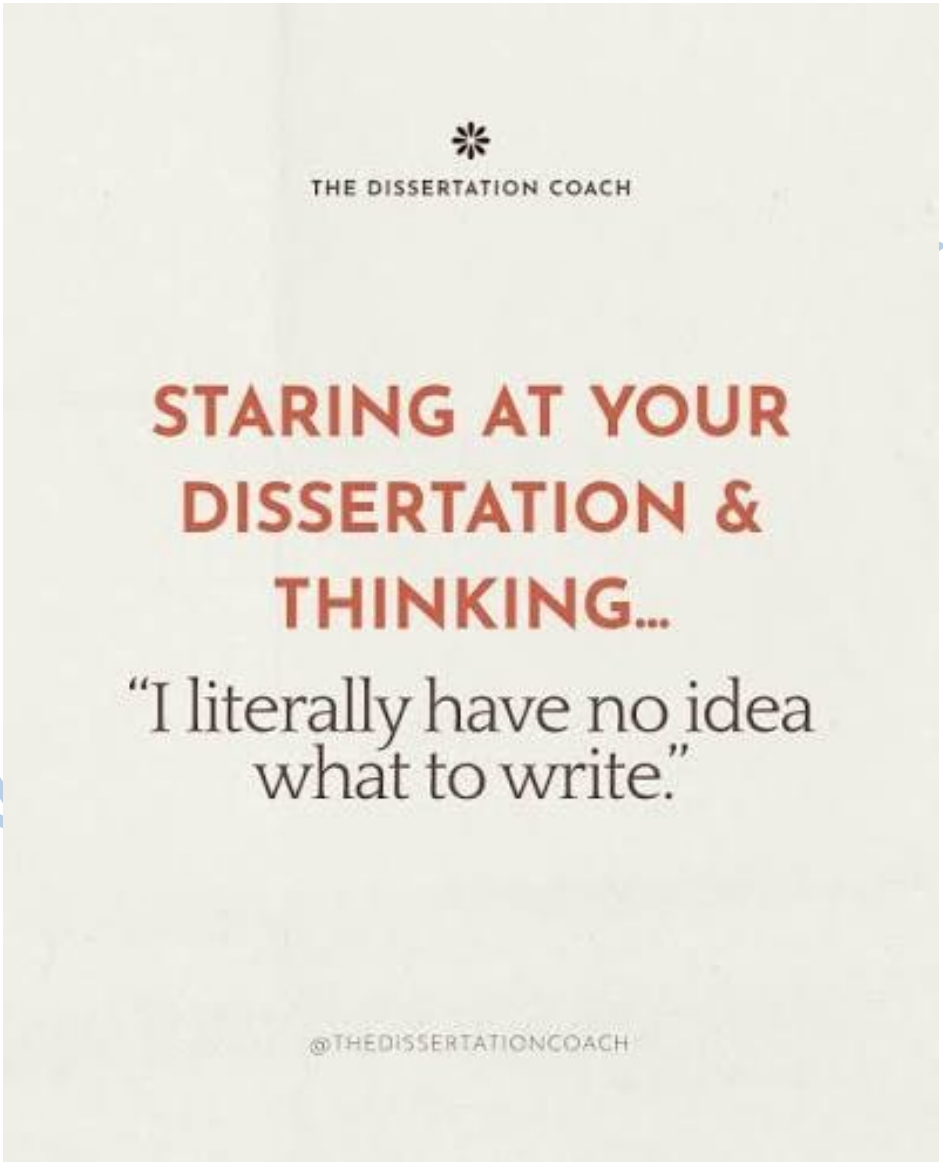
In ONE word, how do you feel about your FYP right now?

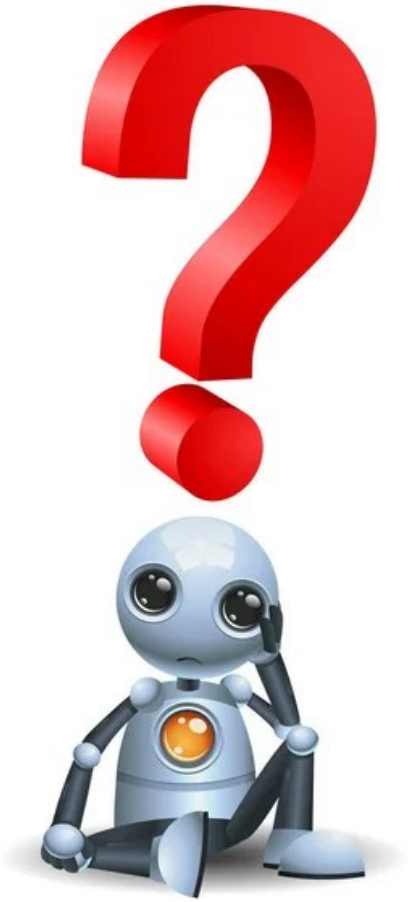
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https://create.vista.com/photos/thinking-robot-isolated/

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OUTLINE

1. What is Dissertation and what it should include
2. Ideal Thesis Structure
3. Abstract
4. Chapter 1: Introduction
5. Chapter 2: Literature Review and/or Theory
6. Chapter 3: Methodology/Project Work
7. Chapter 4: Results and Discussion
8. Chapter 5: Conclusion and Recommendation
9. References
10. Appendices



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What is Dissertation and what it should include?

A dissertation is a **structured academic document** that presents the **complete research/project journey** from **problem identification to conclusions and contributions**.

Front Matter

1. Title Page
2. Certification
3. Abstract
4. Acknowledgements
5. Table of Contents
6. List of Figures
7. List of Tables
8. Abbreviations and Nomenclatures

Chapter 1-5

9. Chapter 1: Introduction
10. Chapter 2: Literature Review and/or Theory
11. Chapter 3: Methodology/Project Work
12. Chapter 4: Results and Discussion
13. Chapter 5: Conclusion and Recommendation

Back Matter

14. References
15. Appendices



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Ideal Thesis Structure

CHAPTER 1 INTRODUCTION	CHAPTER 2 LITERATURE REVIEW	CHAPTER 3 RESEARCH METHODOLOGY	CHAPTER 4 RESULTS AND DISCUSSIONS	CHAPTER 5 CONCLUSION AND FUTURE WORK
• *Why this research?* • *What problem are you solving?*	• *What others have done in the past?* *and* • *What are the limitations in exiting research?*	• *How will you solve the problem?* • *Steps/Methods/ Framework/Flow*	• *What you found as an outcome of your research,* *and* • *What does it mean?*	• *What you achieved?* *and* • *what can be done next?*
Problem → **Objectives**	**What others did?** → **Research gap**	**How you solved it?** → **Your Approach**	**What you found?** → **Meaning**	**Summary/Contribution** → **Future**

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Abstract

YES

- Write last
- Summarise the study
  - Background
  - Problem
  - Objectives
  - Method
  - Findings
  - Contribution
- Keep ≤ 1 page
- Use one continuous paragraph
- Highlight key findings
- State contribution clearly
- Use clear, simple language

NO

AVOID THESE COMMON MISTAKES

- Citations/references
- Tables, figures, equations, bullets
- Multiple paragraphs
- New information
- Excessive background
- Too much methodological detail
- Vague statements (e.g., "Results are discussed")

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CHAPTER 1
INTRODUCTION

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Chapter 1 - Introduction

Purpose of Introduction Chapter (Chapter 1)

- **Sets the stage for the whole thesis**
  - Chapter 1 is like the ***opening scene of a movie***
    - by introducing the topic, sets the context, and helps readers understand why the study matters.
- **Explains background, problem, and objectives**
  - Background → gives general context of the research.
  - Problem → identifies the specific issue that needs to be solved.
  - Objectives → what you need to achieve to address the problem.
- **Shows why the research is important**
  - Justifies the significance of the study.
    - without this, the thesis may look like it has no purpose.
- **Provides roadmap for the reader**
  - Chapter 1 usually ends with a short "*thesis structure*" section.
  - This helps examiners, supervisors, or any reader to navigate the thesis more easily.



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Major sections of Chapter 1



Major sections of Chapter 1

1.1 Overview



- The Overview section introduces the chapter by briefly explaining its purpose, key topics, and organization.
- It provides readers with a roadmap of what will be discussed in the chapter before presenting the detailed content.



Example: This chapter introduces the study on machine learning for predicting sinkhole formation. It presents the research background (Section [.....]), problem statement (Section [.....]), research objectives (Section [.....]), and research questions (Section [.....]). The scope and significance of the study are also discussed in Section [.....], followed by an outline of the thesis structure in Section [.....]. This chapter provides the foundation for understanding the motivation and direction of the research.

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Major sections of Chapter 1

1.2 Research Background



- **General Context:** Start broad. Describe the field of study (e.g., agriculture, education, healthcare, computer science).
- **Current Trends/Developments:** Talk about what's happening in this area, new technologies, policies, or global issues.
- **Existing Knowledge/Practices:** Summarize what researchers are already doing in this domain.
- **Challenges/Gaps Emerging:** Narrow down focus to highlight problems that remain unsolved.
- **Lead to the Research Problem:** End by linking this background to why your specific study is needed.

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Major sections of Chapter 1

1.3 Problem Statement



- ***Clearly define the gap*** or issue in current knowledge, practice, or technology.
- ***Show why the problem*** is important and worth solving.
- Explain what has ***not been done***, what is ***missing***, or what ***challenges remain***.



Example: *"Despite advances in geological monitoring, existing models still struggle to accurately predict sinkhole formation, leading to unreliable risk assessments (Smith & Johnson, 2025)."*



Support the problem statement with **relevant citations** to justify it.

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Major sections of Chapter 1

1.4 Research Questions



- List the ***specific questions*** your study seeks to answer.
- They should ***directly be linked*** to the ***problem statement***.
- Start with "***What***", "***How***" or "***To what extent***" to make them ***precise***.
- Usually ***2–4 focused questions***.

Example:



i. What factors influence sinkhole formation?

ii. How can ML enhance the accuracy of sinkhole prediction?

iii. To what extent can ML improve early warning and risk assessment for sinkholes?

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Major sections of Chapter 1

1.5 Aim of the Study



- Write the overall **big picture** of your research.
- One to two sentences, **broad but focused**.

Example:



*"The aim of this study is **to develop a ML model** that **improves the prediction of sinkhole formation** by analyzing **environmental and geological data.**"*

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Major sections of Chapter 1

1.6 Research Objectives

- **Link to Research Questions:** Each objective should answer at least one question.
- **Specific & Measurable:** Use clear action verbs (analyze, design, develop, evaluate).
- **Bullet Points:** Keep them concise and easy to read.



Example:



i. To investigate the limitations of existing ML models in predicting factors influencing sinkhole formation.
ii. To develop a ML model for accurate sinkhole prediction.
iii. To evaluate and compare the model's performance with existing prediction approaches.

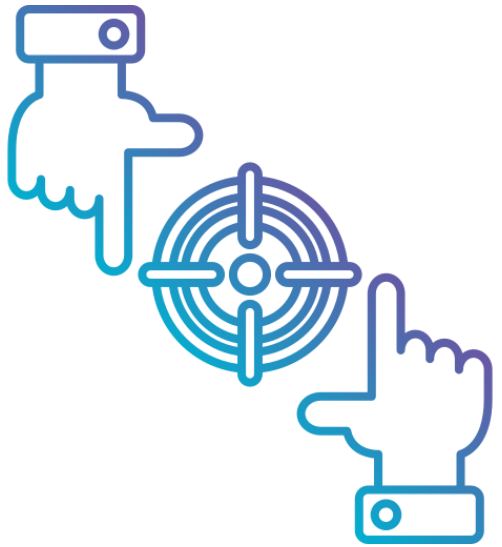
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*Treat objectives as **milestones**, aim for **3-4**, and ensure they're **feasible within your time/resources**.*

Major sections of Chapter 1

1.7 Scope of Study



- Define the ***boundaries of your research***.
- Clarify what is ***included and excluded***.
- Prevents your study from being ***"too broad."***

Example:



- *"This study focuses on using ML to predict sinkhole formation in **Malaysia only**."*
- *"The research analyzes environmental and geological data, **excluding socio-economic factors**."*
- *"The study evaluates model accuracy and performance but **does not cover real-time field deployment**."*

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Major sections of Chapter 1

1.8 Significance of the Research



- Explain ***why*** your ***study is important***.
- ***Who*** will ***benefit*** (academics, industry, policymakers, farmers, etc.)?
- Highlight ***contributions*** to knowledge and practice.

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Example:



- *"This research will support engineers and policymakers in making data-driven decisions, improve early sinkhole risk detection, and contribute to safer infrastructure planning and disaster prevention."*

Major sections of Chapter 1

1.9 Thesis Structure



- Provide a ***roadmap of the thesis*** chapters.
- Briefly describe ***what each chapter covers***.

Example:



- *Chapter 1 introduces the research problem, objectives, and scope.*
- *Chapter 2 reviews relevant literature.*
- *Chapter 3 explains the methodology.*
- *Chapter 4 presents results and discussion.*
- *Chapter 5 concludes and suggests future work.*

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Major sections of Chapter 1

1.8 Thesis Structure



Example Figure

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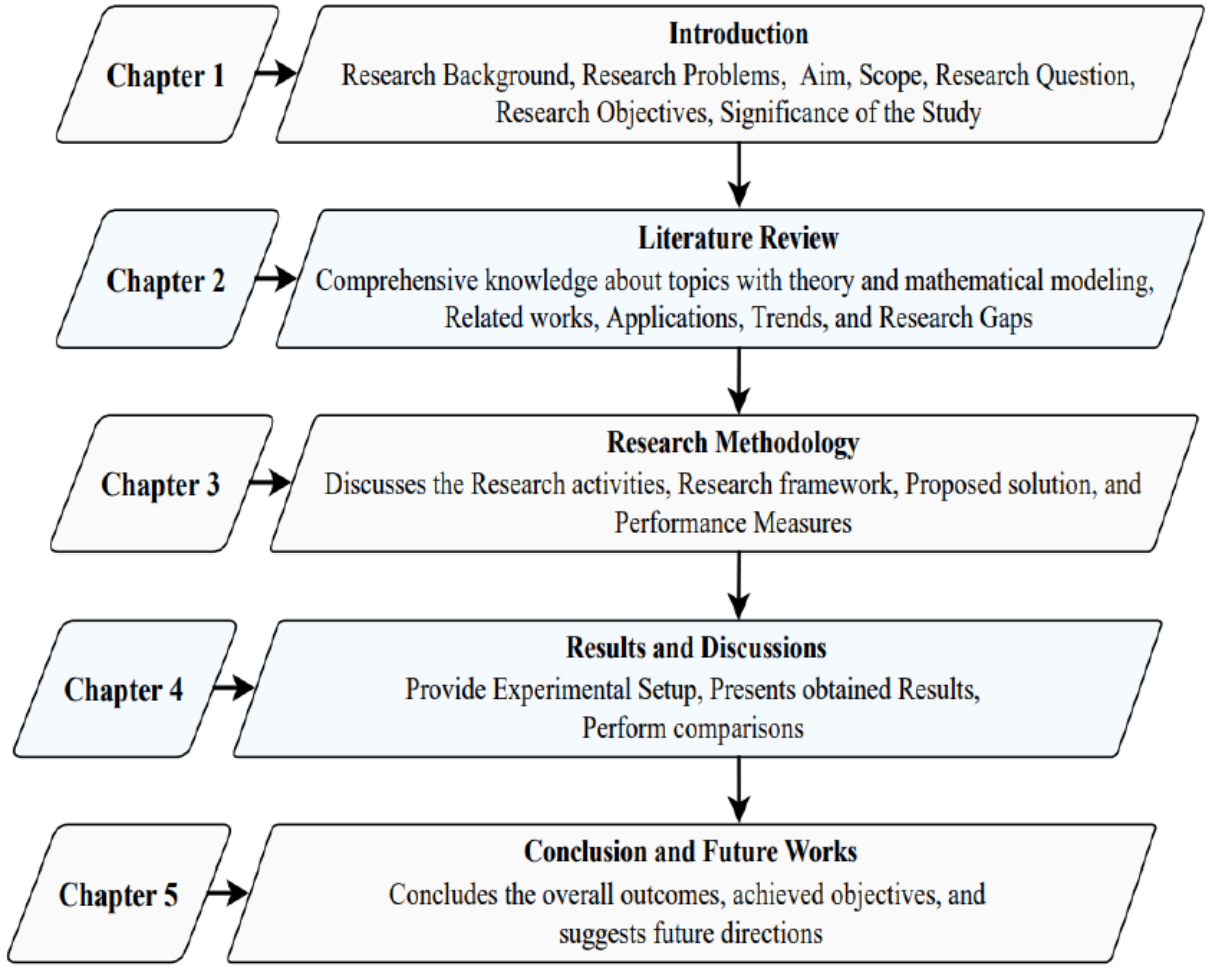


Figure 1.3: Thesis structure

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Do's and Don'ts checklist for writing Chapter One

Do's (What to follow)	Don'ts (What to avoid)
• Be **clear and concise** in your introduction.	• **Don't overload** with unnecessary background details.
• **Clearly define** the problem statement, objectives and research questions directly.	• Don't be **vague** about the problem, objectives, or questions.
• **Highlight the significance** and **contribution** of your study.	• Don't skip justification of why your study is important.
• **Maintain logical flow** from problem → objectives → significance → scope.	• **Don't ignore** structure/present ideas in a scattered way.
• **Set clear boundaries** by defining scope and limitations.	• **Don't exaggerate or overpromise** the scope of your study.
• Use professional language.	• Avoid repetitive words.



DO'S



DON'TS

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Purpose of a Literature Review



- Reviews **past studies related to your research topic**.
- Defines **key concepts and theories**.
- Explores **who studied what, why, and what they discovered**.
- Helps you **understand existing research**, **refine research questions**, and **identify research gaps**.
- Supports the development of your **hypotheses or research direction**.
- Demonstrates your **knowledge and understanding of the field**.

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Steps to search studies for Literature Review Chapter



i. Search relevant literature aligned with your topic

ii. Analyze & evaluate the importance of paper

iii. Create an excel to list the important studies

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Steps to search studies for Literature Review Chapter

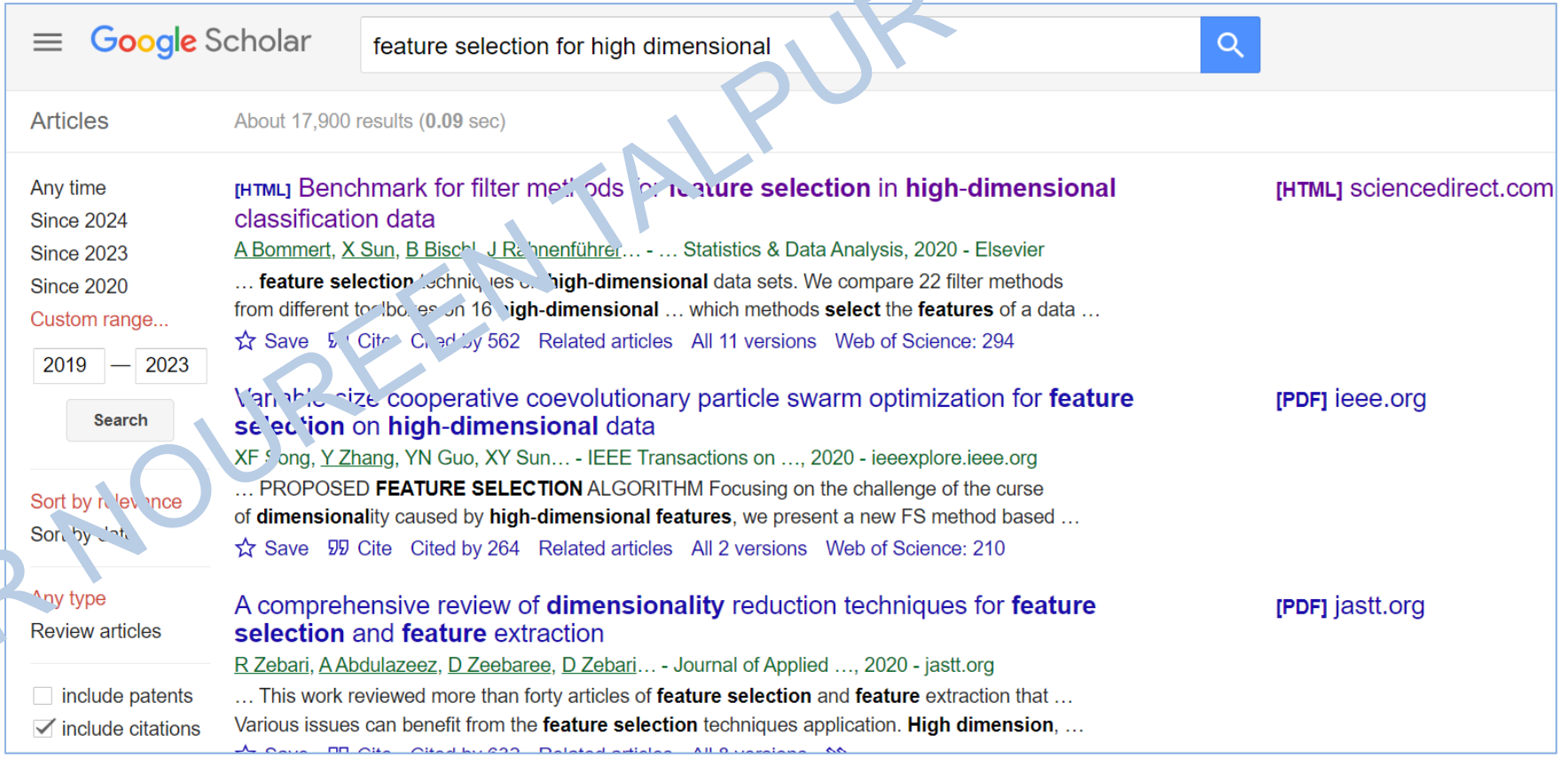
i - Search relevant literature aligned with your topic



Google Scholar

https://scholar.google.com/

*Find most relevant studies from **past 5 years***

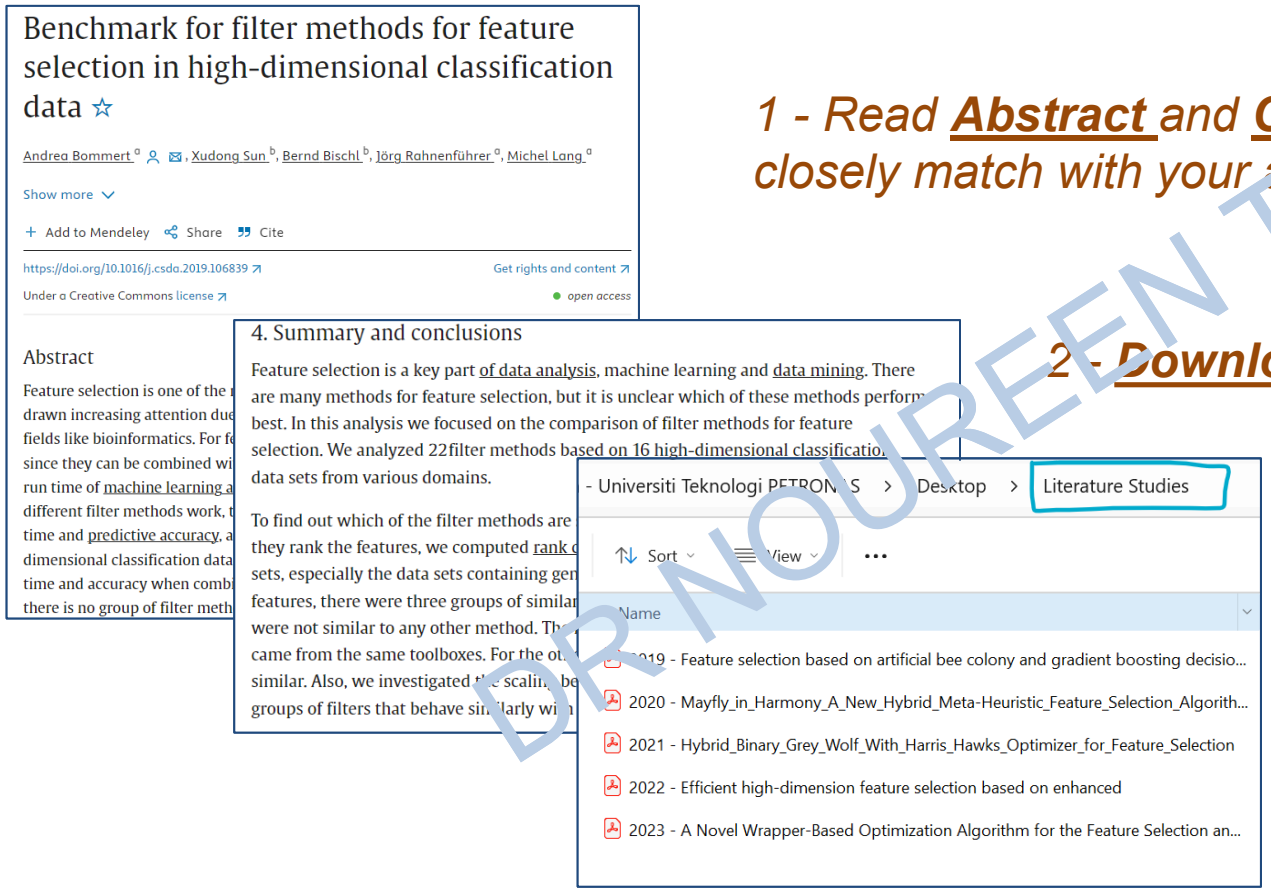


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Steps to search studies for Literature Review Chapter

ii - Analyze & evaluate the importance of paper



*1 - Read **Abstract** and **Conclusion** of the study to evaluate if the study closely match with your area of the study*

*2 - **Download the paper** if its relevant to read details later*

*3 - **Create a folder** and **save** all those studies **organized by year of study***

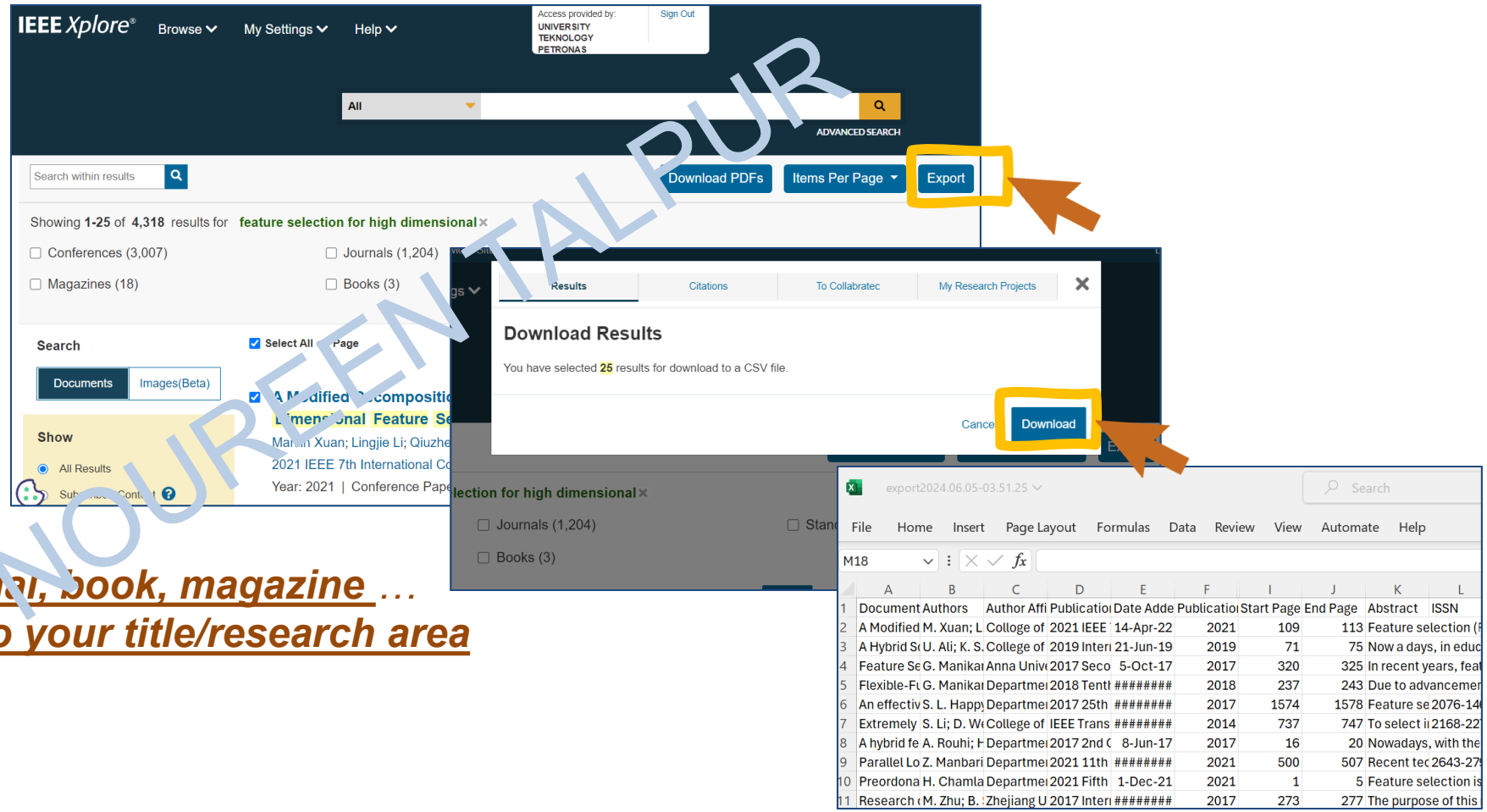
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Steps to search studies for Literature Review Chapter

iii - Create an excel to list the important studies

IEEE Xplore® Digital Library

https://ieeexplore.ieee.org/Xplore/home.jsp



1. ***Search*** *by topic*
2. *Choose by* ***Conference, journal, book, magazine*** *…*
3. *Select the studies* ***matching to your title/research area***
4. ***Export*** *the list in CSV*

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Other resources to find literature:

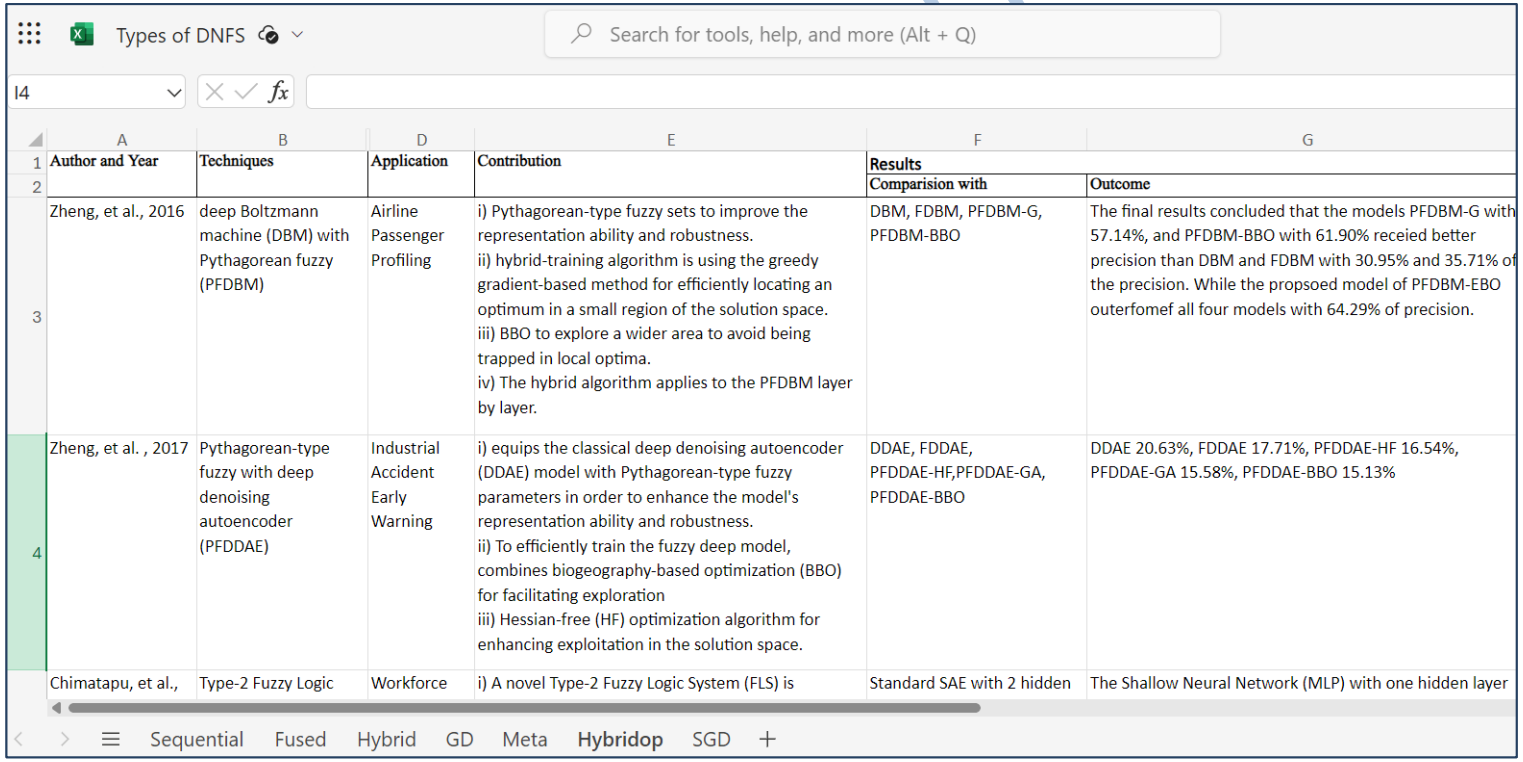
- https://link.springer.com/
- https://www.sciencedirect.com/

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Steps to search studies for Literature Review Chapter

iii - Create an excel to list of the important studies

Create your own Excel file of the studies that are most relevant, and you want to add into your thesis



	A	B	D	E	F	G
1	Author and Year	Techniques	Application	Contribution	Results	
2					Comparision with	Outcome
3	Zheng, et al., 2016	deep Boltzmann machine (DBM) with Pythagorean fuzzy (PFDBM)	Airline Passenger Profiling	i) Pythagorean-type fuzzy sets to improve the representation ability and robustness. ii) hybrid-training algorithm is using the greedy gradient-based method for efficiently locating an optimum in a small region of the solution space. iii) BBO to explore a wider area to avoid being trapped in local optima. iv) The hybrid algorithm applies to the PFDBM layer by layer.	DBM, FDBM, PFDBM-G, PFDBM-BBO	The final results concluded that the models PFDBM-G with 57.14%, and PFDBM-BBO with 61.90% receied better precision than DBM and FDBM with 30.95% and 35.71% of the precision. While the propsoed model of PFDBM-EBO outerfomef all four models with 64.29% of precision.
4	Zheng, et al. , 2017	Pythagorean-type fuzzy with deep denoising autoencoder (PFDDAE)	Industrial Accident Early Warning	i) equips the classical deep denoising autoencoder (DDAE) model with Pythagorean-type fuzzy parameters in order to enhance the model's representation ability and robustness. ii) To efficiently train the fuzzy deep model, combines biogeography-based optimization (BBO) for facilitating exploration iii) Hessian-free (HF) optimization algorithm for enhancing exploitation in the solution space.	DDAE, FDDAE, PFDDAE-HF,PFDDAE-GA, PFDDAE-BBO	DDAE 20.63%, FDDAE 17.71%, PFDDAE-HF 16.54%, PFDDAE-GA 15.58%, PFDDAE-BBO 15.13%
	Chimatapu, et al.,	Type-2 Fuzzy Logic	Workforce	i) A novel Type-2 Fuzzy Logic System (FLS) is	Standard SAE with 2 hidden	The Shallow Neural Network (MLP) with one hidden layer

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Writing Literature Review

Make an outline of the Chapter 2 – Important Sections

2.1 Overview	• Overview ***Purpose and Importance*** of the Literature Review • Explanation of the ***Structure and Organization*** of the Chapter
2.2 Theoretical Framework	• ***Introduction*** to ***Theoretical Frameworks*** relevant to the research topic • Explanation of ***key concepts*** and constructs
2.3 Related Works	• ***Exploration of historical background*** of Topic. Review of ***recent studies*** and developments in the field - ***Comparison of different approaches and methodologies***
2.4 Research Gap	• Identify ***research gaps/unanswered questions*** in literature. Discuss ***challenges/ limitations*** in past studies. ***Justify why addressing those gaps are important***
2.5 Chapter Summary	• ***Recap of key findings/*****insights** from the literature review • Summary of ***significance of the literature review*** to the overall study

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Writing Literature Review

Summarize literature studies in RELATED WORKS the form of table

Table 1.2: Summary of Comparative Studies in Cancer Diagnosis Using Machine Learning

Author & Study	Method	Application	Comparison with	Metrics	Results
Khanna et al. (2021)	Hybrid (KNN + SVM)	Breast Cancer Prognosis	KNN, SVM	Accuracy	• The hybrid KNN+SVM model 96.5% • 94.2% for SVM. • 93.1% for KNN.
Wang & Zhang (2023)	Hybrid (RF + KNN)	Breast Cancer Risk Prediction	RF, KNN	Accuracy	• The hybrid RF+KNN mod[illegible]achi[illegible]ed a[illegible]curacy of 94.8%.
Gupta et al. (2024)	SVM with RBF Kernel	Breast Cancer Detection	Proposed Hybrid (XGBoost + CNN)	Accuracy	• The S[illegible]M [illegible]odel a[illegible]ieved a robust accuracy of 92.3% [illegible]ropo[illegible]ed hybrid model was outperformed by reached 96.1% accuracy.
Lee & Park (2025)	XGBoost with SHAP	Multi-Cancer (Pan-Cancer) Classification	SVM	Acc[illegible]	• XGBoost provided the best balance of accuracy (95.7%). • It consistently outperformed by XGBoost. • SVM (91.2%) across multiple cancer types.

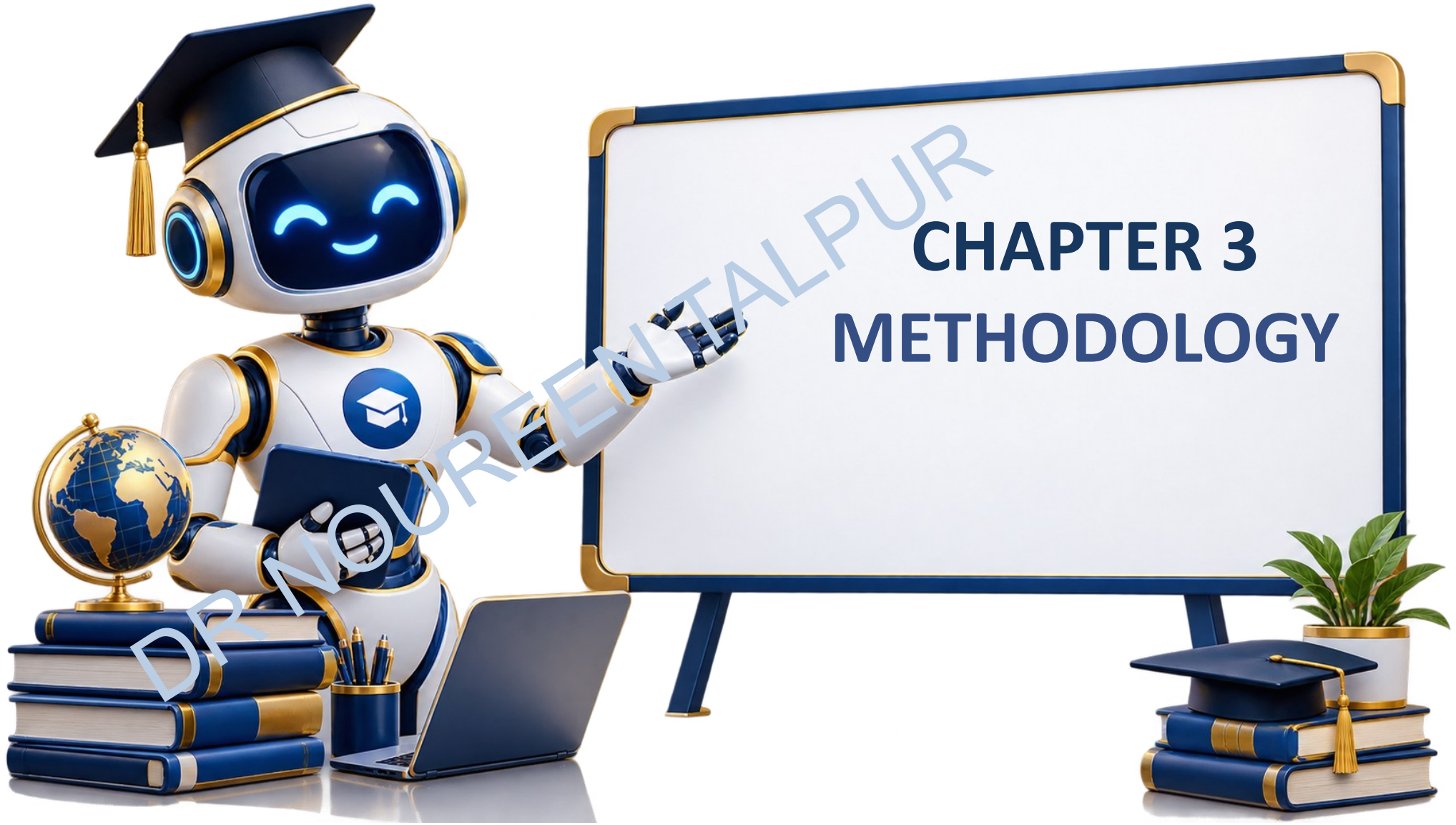
Tabl[illegible] 2: Su[illegible]ary of the recent literature for the task of FS in medical sciences

Study, year	Algo[illegible]thm	Application	Major contributions	Outcomes
[61[illegible] 2019	Crow[illegible]earch Algorithm (CSO)	Lung cancer detection	• Lungs modality is determined using computed tomography (CT) imaging, and classification is done using Probabilistic Neural Networks (PNN). • Feature extraction is done using the gray-level co-occurrence matrix. • CSO is used for the task of FS	The findings show that the CSO-based FS technique successfully delivered 90% accuracy.
[62], 2020	Two-stage Genetic Algorithm-feature Transformation (2-Tra-GA)	Hepatitis prediction	• A 2-Tra-GA-based feature transformation is proposed with two phases of FS and augmentation • The lagrangian SVM (LSVM) approach is used to predict the hepatitis	The novel 2-Tra-GA-LSVM technique showed faster convergence with 90.32% accuracy in distinguishing between survived and deceased patients with hepatitis.
[63], 2021	Krill Herd Algorithm (KHA)	Brain epilepsy seizure detection	• The EEG data is used to identify epileptic seizures using a deep learning approach. • KHA identifies the most optimal features from the brain signal data. • General Adversarial Networks classify the selected features.	The KNN classifier detects the seizure with 99.25% when compared to the baseline method owing to the efficient FS mechanisms of the KHA algorithm.
[64], 2021	Grasshopper Optimization Algorithm (GOA)	Diabetes disease diagnosis	• GOA is used to select the key features from diabetes type II data. • Three classifiers including SVM, NB, and Tree are used to classify the features of the diabetes data.	Compared to the NB and Tree classifier, SVM-GOA showed the best results having the least mean squared error of 3.0351.

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Put this table at the end of related works Section after discussing related works from past 5 years

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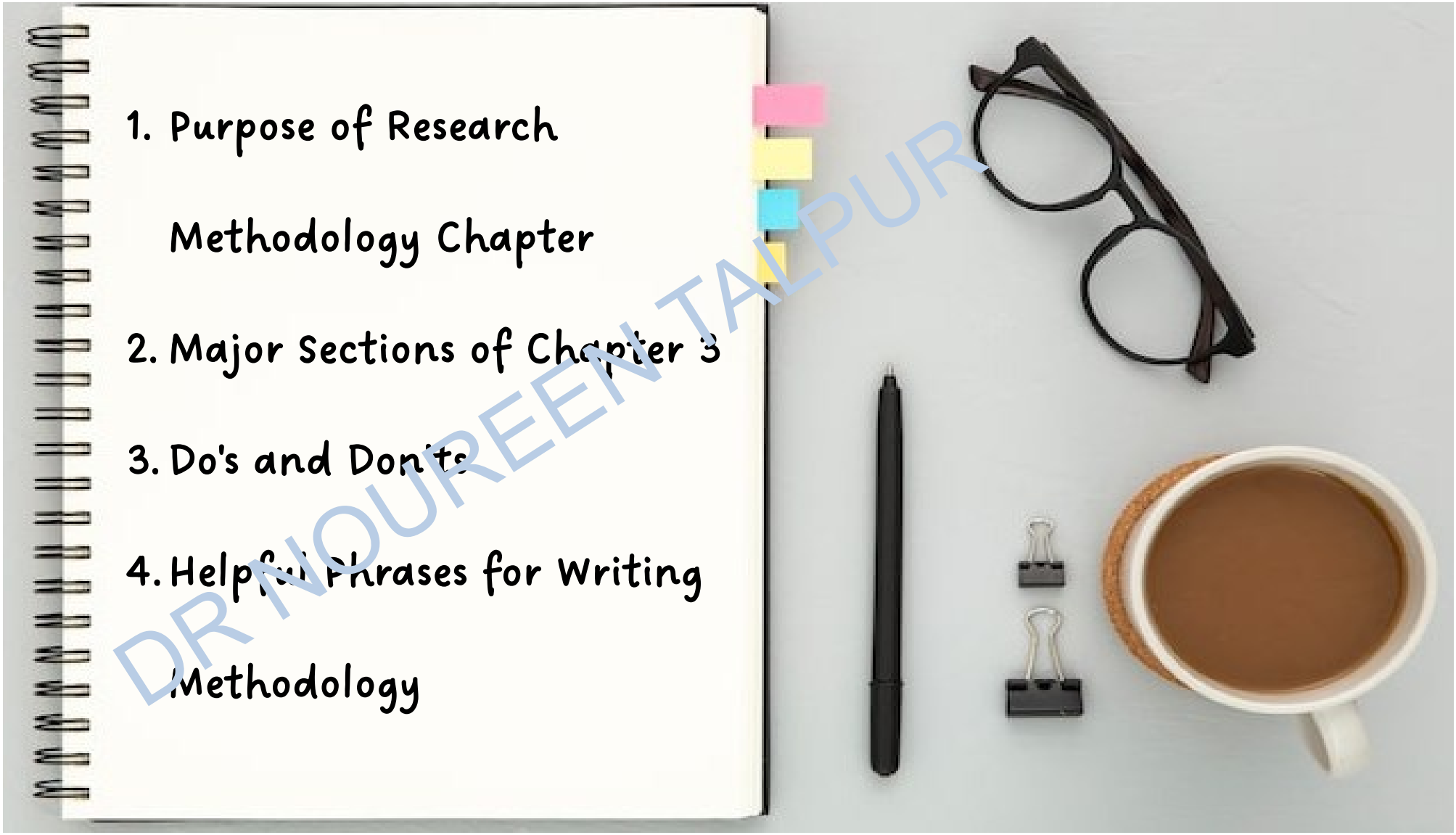


CHAPTER 3
METHODOLOGY

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OUTLINE



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Purpose of Research Methodology Chapter



Sets the foundation for conducting the study

- **Explains** how the research was carried out.
- **Describes** methods used to achieve research objectives.
- **Justifies** why particular approaches were selected.
- **Demonstrates** the reliability and validity of the study.

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Provides transparency

A reader should be able to understand:

- What was done?
- How was it done?
- Why was it done this way?



Key Question Answered

"How did you conduct the research to achieve your objectives?"

Major Sections of Chapter 3

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For Machine Learning–Based Research

3.1 Overview
3.2 Proposed Methodology
3.3 Tools and Techniques
3.4 Chapter Summary



For Software, Application, or Hardware Development

3.1 Overview
3.2 Project Framework
3.3 System Design and Architecture
3.4 Hardware Design
  3.4.1 Components List
3.5 Software Development
3.6 System Integration and Testing
3.7 Chapter Summary



The structure of Chapter 3 should be adapted to the nature of the project.

- **Machine learning research** focuses on the **proposed methodology, tools, techniques, dataset, model development, and evaluation**.
- **Development-based projects** focus on the **project framework, system architecture, hardware or software design, integration, and testing.**

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Major Sections of Chapter 3

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For Machine Learning–Based Research

3.1 Overview
3.2 Research Methodology
3.3 Tools and Techniques
3.4 Chapter Summary



Section	What to write
3.1 Overview	Research workflow, methodology description, key sections
3.2 Proposed Methodology	CRISP-DM, KDD, SEMMA or your own methodology
3.3 Tools & Techniques	Python, ML libraries, algorithms, evaluation methods
3.4 Chapter Summary	Recap methodology and prepare for Chapter 4

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This is a suggested structure only. Chapter 3 sections may be added, removed, renamed, or reorganized based on the project's methodology and requirements.

Major Sections of Chapter 3

3.2 Research Methodology

What is Research Methodology?

A research methodology provides a **visual representation** of the overall research/project process.



Think of it as a

"ROADMAP"

of the entire project.

WHY IS IT IMPORTANT?

- **Shows** the relationship among research activities.
- **Helps** readers understand the study at a glance.
- **Provides** logical flow of the research process.



Every component shown in the framework **must be explained** in the chapter.

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Major Sections of Chapter 3

RESEARCH FLOW EXAMPLES FOR ML RELATED PROJECTS

3.2 Research Methodology

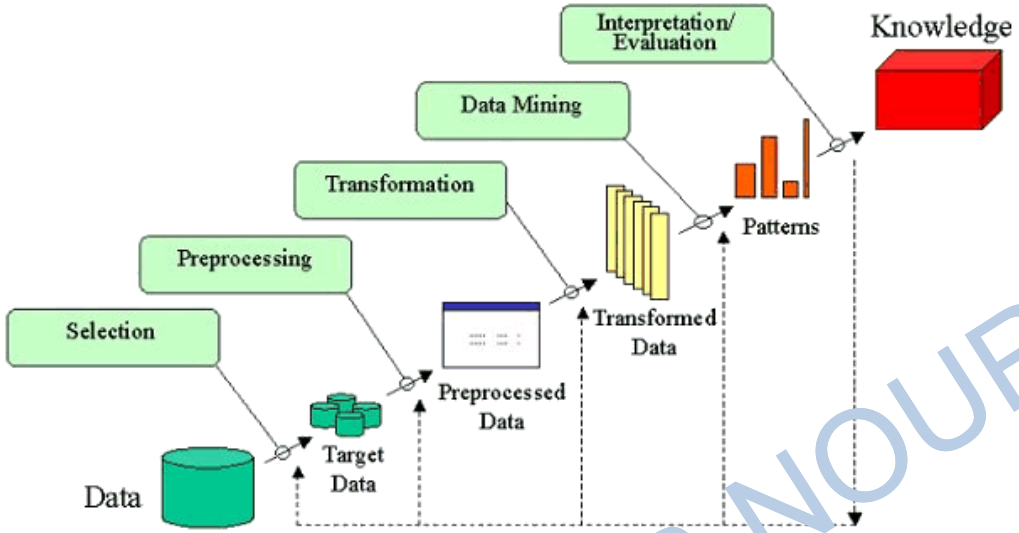
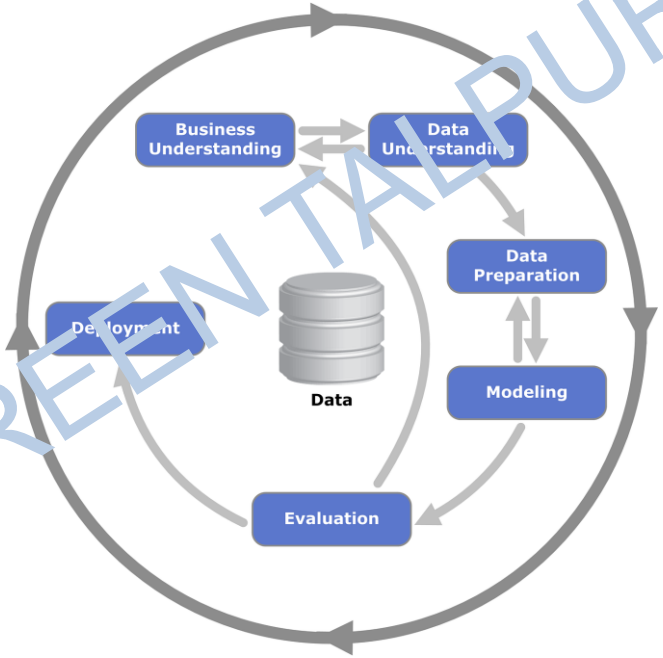
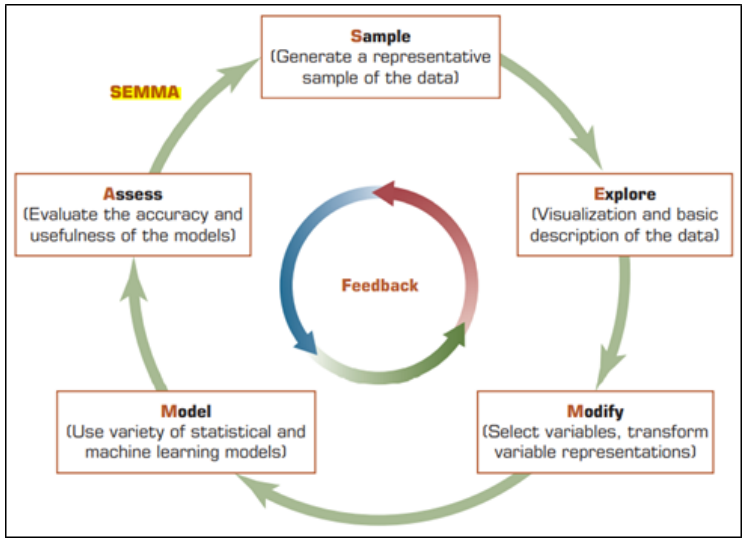


Diagram of KDD Model. (Source)



CRISP - DM



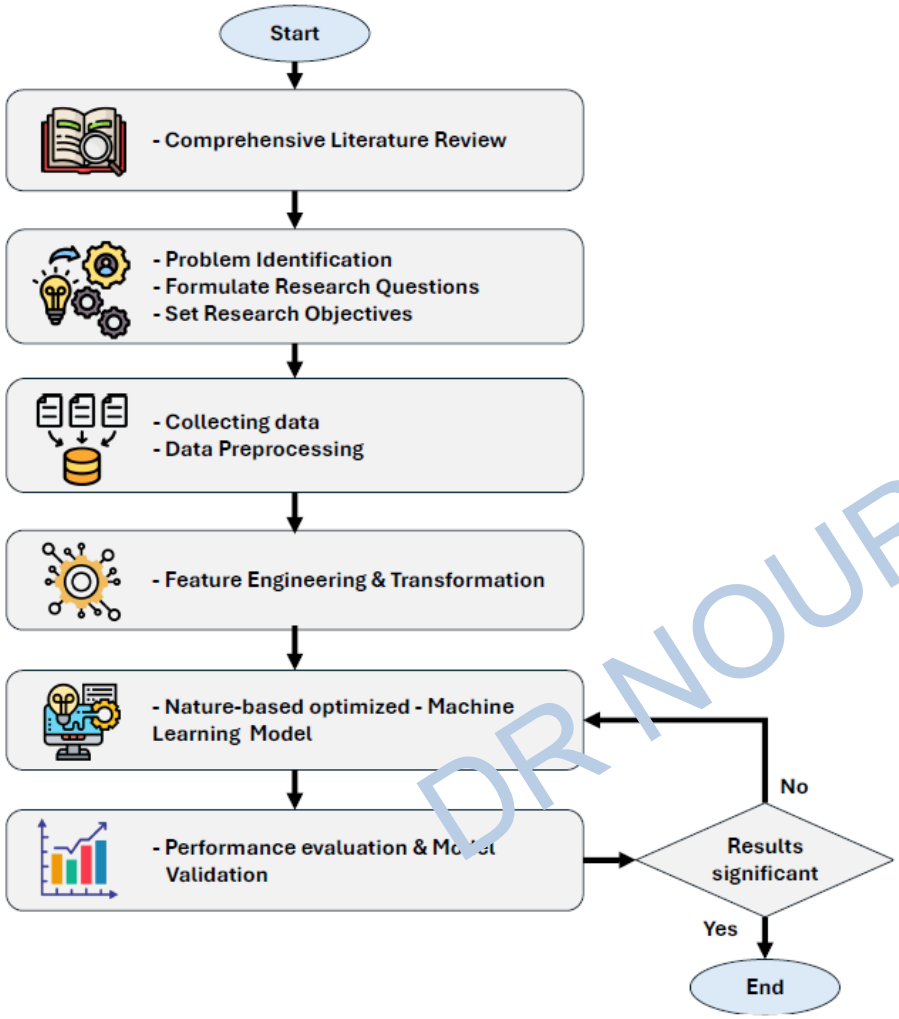
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Major Sections of Chapter 3

RESEARCH FLOW EXAMPLES FOR ML RELATED PROJECTS

3.2 Research Methodology



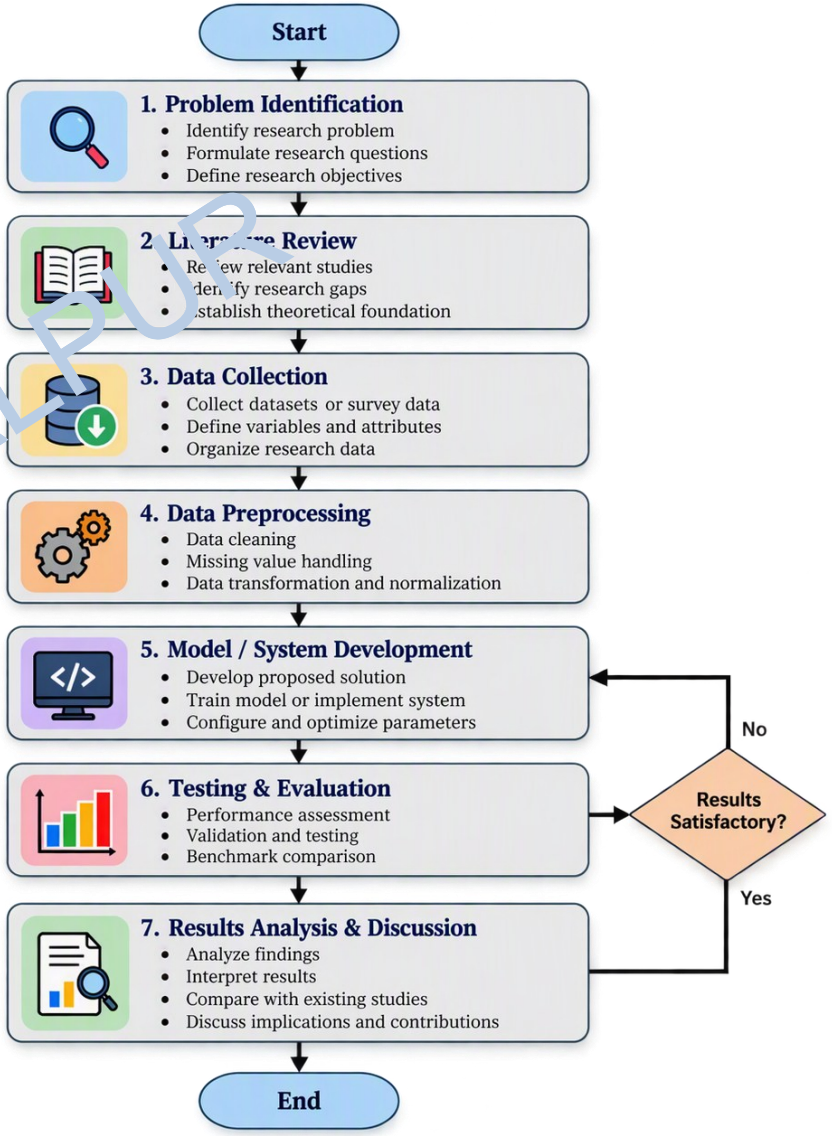


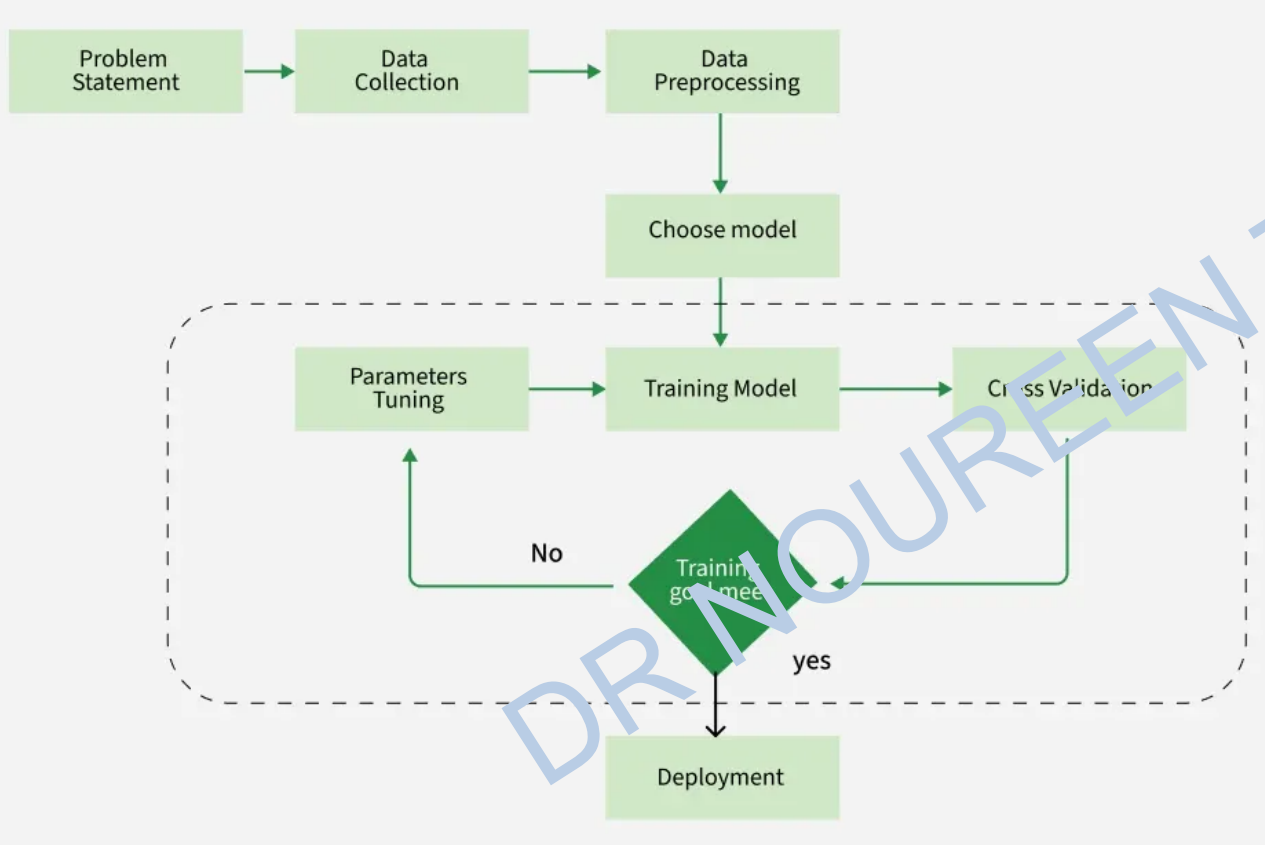
Figure 3.1: Research Methodology Framework

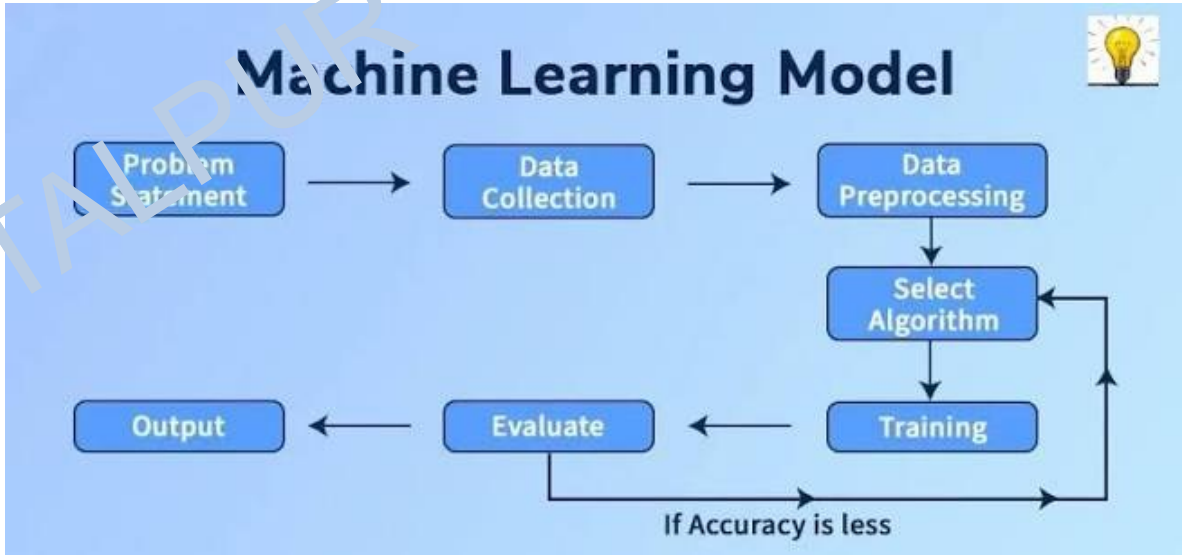
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Major Sections of Chapter 3

RESEARCH FLOW EXAMPLES FOR ML RELATED PROJECTS

3.2 Research Methodology





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Major Sections of Chapter 3

3.3 Tools and Technologies

Clearly mention all software and technologies used.



IMPORTANT

Always state SOFTWARE VERSIONS where possible.

Purpose	Tool / Technology
Programming	Python 3.12
Analysis	SPSS 29
Visualization	Power BI
Database	MySQL
Development	VS Code
Machine Learning	Scikit-learn
Deep Learning	TensorFlow / PyTorch
Version Control	Git / GitHub
Cloud Platform	Google Colab / Kaggle
Computational Environment	Intel i7, 16GB RAM, NVIDIA RTX 4060

Example Declaration Sentence:

Several software tools, technologies, and computational resources were utilized throughout the study. Python 3.12 was used for data processing and programming, while Scikit-learn 1.5 was employed to develop and evaluate the machine learning models. SPSS 29 was used for statistical analysis, and MySQL 8.0 was utilized for database management. Visual Studio Code served as the primary development environment, whereas Power BI was used for data visualization and presentation of the findings.

The computational experiments were conducted using a computer equipped with [processor/model], [RAM] GB of RAM, and [GPU model, if applicable], providing the computational resources required for data processing and model training.

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Major Sections of Chapter 3

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For Software, Application, or Hardware Development

- 3.1 Overview
- 3.2 Project Framework
- 3.3 System Design and Architecture
- 3.4 Hardware Design
  - 3.4.1 Components List
- 3.5 Software Development
- 3.6 System Integration and Testing
- 3.7 Chapter Summary



Section	What to write
3.1 Overview	Development workflow & chapter structure
3.2 Project Framework	SDLC, Agile, Waterfall, etc.
3.3 System Design	Architecture, UML, database, flowcharts
3.4 Hardware Design	Hardware setup & components
3.5 Software Development	Tools, modules, implementation
3.6 Integration & Testing	Testing procedures & validation
3.7 Chapter Summary	Recap methodology and prepare for Chapter 4



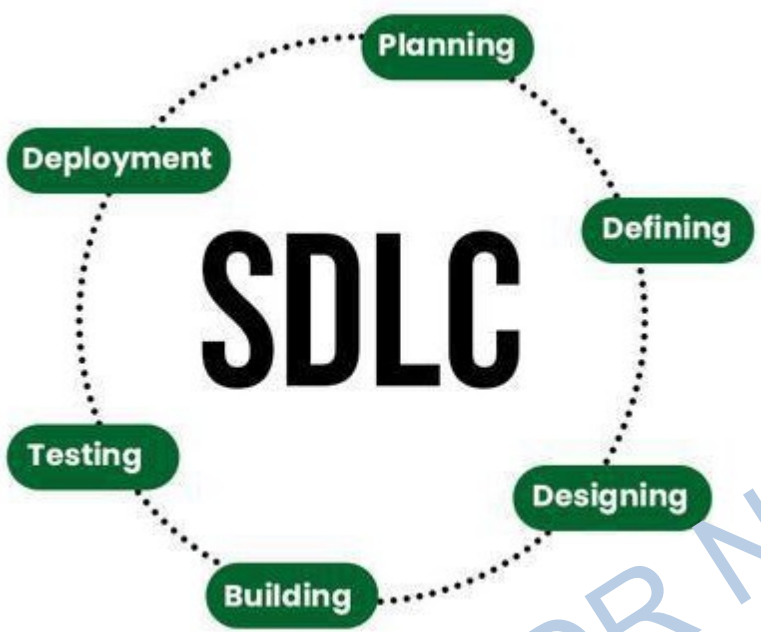
This is a suggested structure only. Chapter 3 sections may be added, removed, renamed, or reorganized based on the project's methodology and requirements.

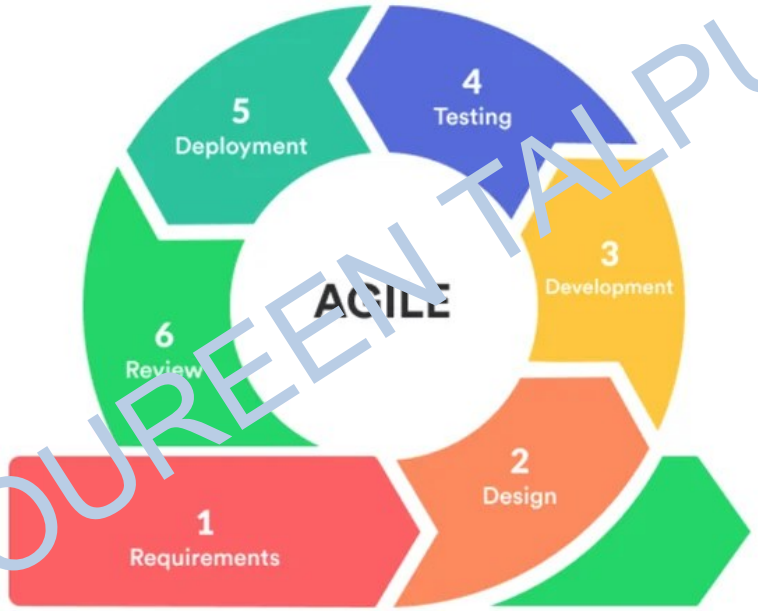
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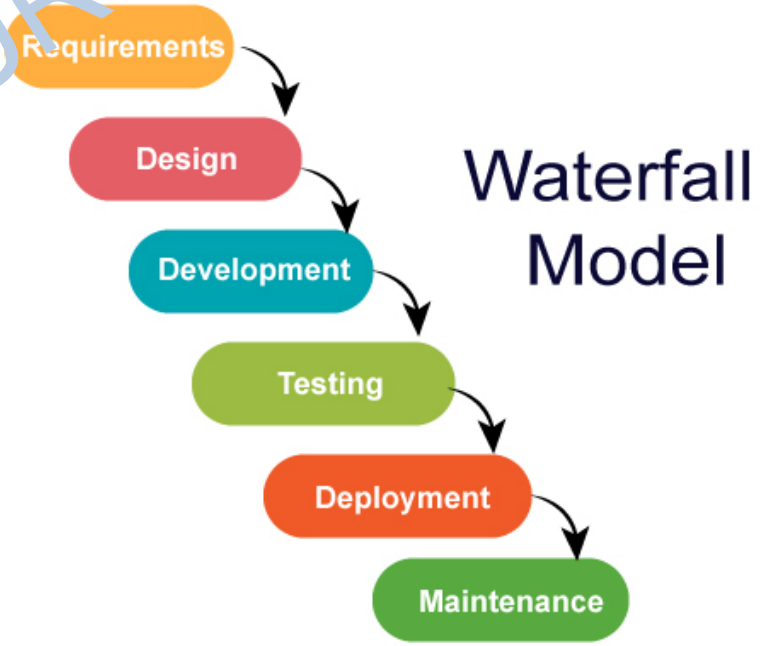
Major Sections of Chapter 3

USE CASE DIAGRAM EXAMPLES FOR APP/SOFTWARE/WEB DEVELOPMENT

3.2 Research Methodology





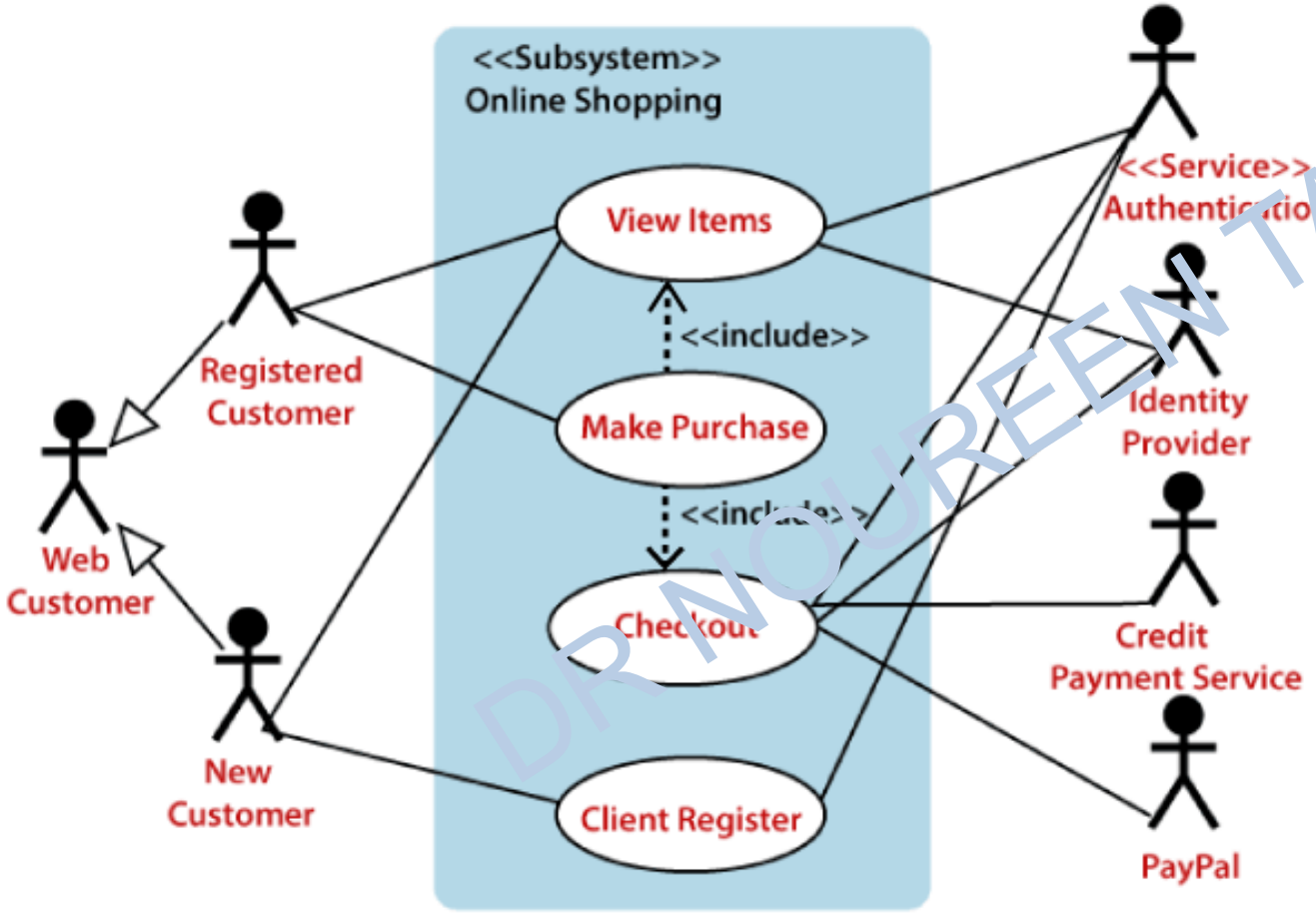


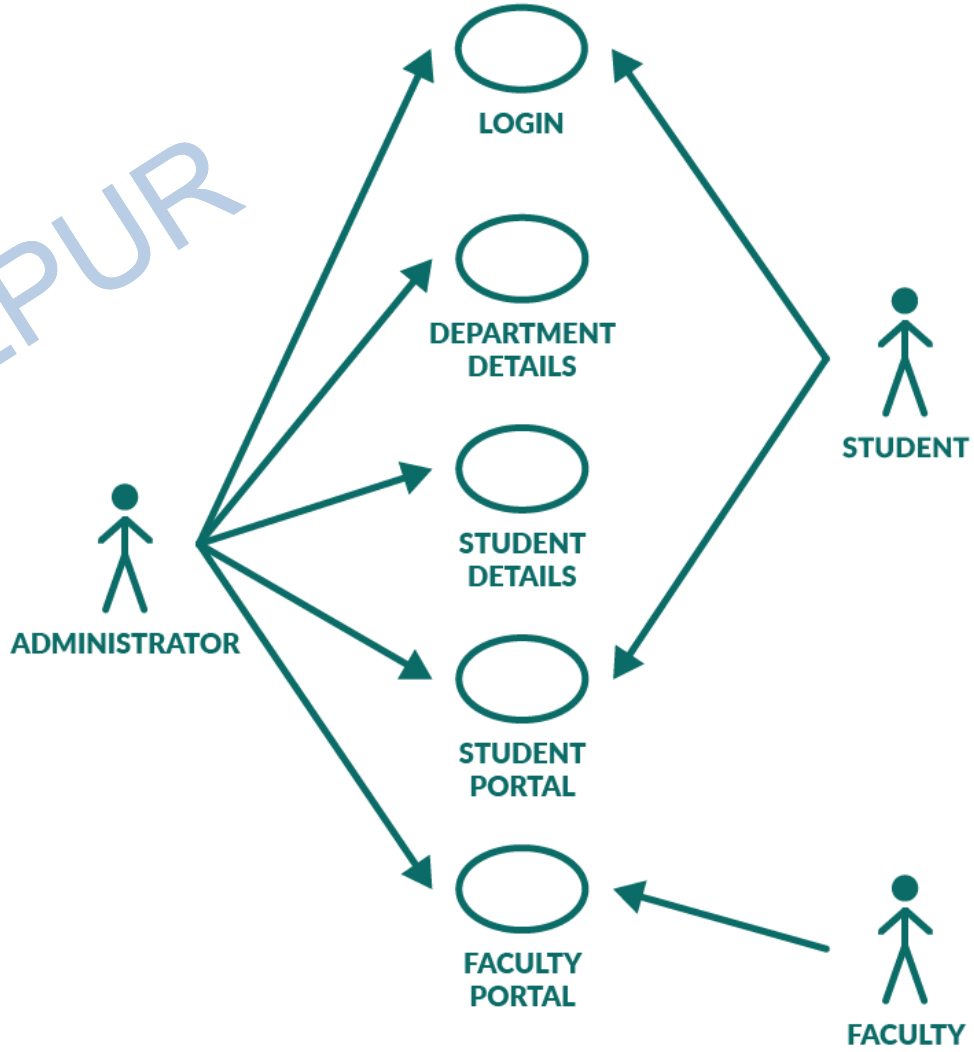
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Major Sections of Chapter 3

USE CASE DIAGRAM EXAMPLES FOR APP/SOFTWARE/WEB DEVELOPMENT

3.2 Research Methodology





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Major Sections of Chapter 3

Data Collection

What is DATA COLLECTION?
The process of **gathering information** required for the study.

Two Major Sources

Primary Data
Collected **directly by the researcher**.

Examples:
- Questionnaire
- Interviews
- Focus Groups
- Observation

Secondary Data
Collected from **existing sources**.

Examples:
- Kaggle
- UCI Repository
- Government Databases
- Company Records
- Published Research Datasets

Kaggle: https://www.kaggle.com/datasets
UCI: https://archive.ics.uci.edu/

Data Collection Example

1. Machine Learning Project
Dataset Source:
- Kaggle
- UCI Machine Learning Repository
- Government Open Data Portal

2. Survey-Based Project
Data Collection Instrument:
- Google Forms
- Microsoft Forms
- Printed Questionnaires

Include:
- Number of participants
- Collection period
- Sampling technique

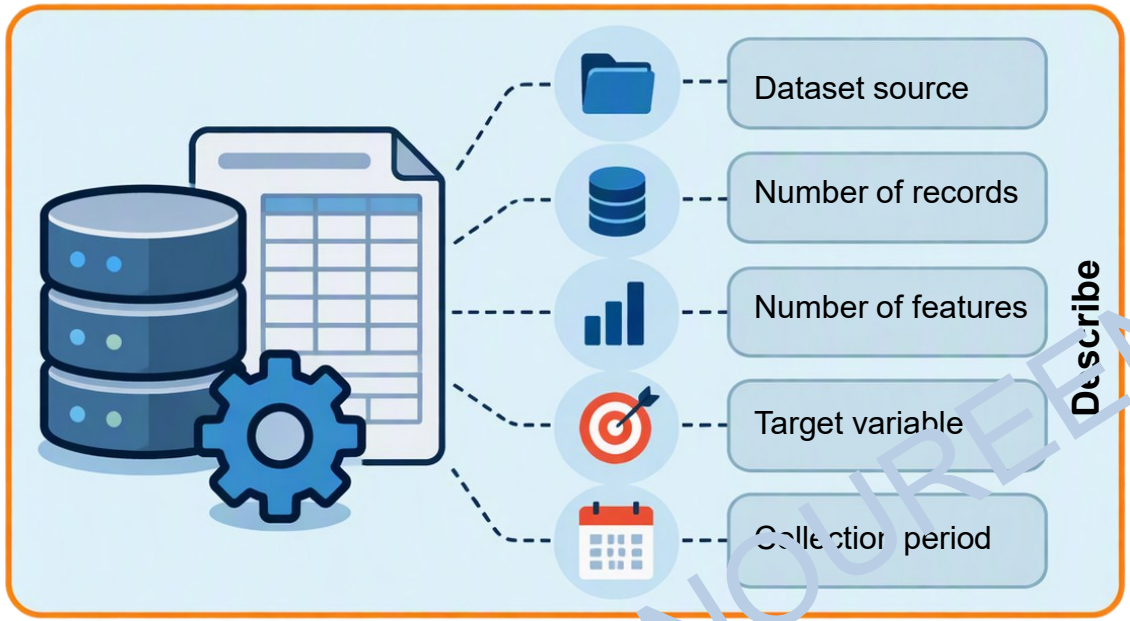


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Major Sections of Chapter 3

Data Collection

FOR MACHINE LEARNING STUDIES



Dataset Description Example - Example Paragraph

The study utilizes the Heart Disease dataset obtained from the UCI Machine Learning Repository. The dataset contains 1025 patient records and 14 attributes, including age, cholesterol level, blood pressure, and heart rate. The target variable indicates whether the patient has heart disease.

FOR SURVEY-BASED STUDIES



Dataset Description Example - Example Paragraph

The study collects data from UTP undergraduate students through a structured questionnaire. A total of 120 respondents are selected using convenience sampling. Participants are undergraduate students enrolled at UTP who meet the specified participation criteria. The questionnaire measures students' perceptions and acceptance of AI learning tools using a Likert-scale format.

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Major Sections of Chapter 3

Chapter 3 Writing Tips

Every Objective Must Be Reflected

Objective 1 → Method Step
Objective 2 → Method Step
Objective 3 → Method Step



COMMON MISTAKE

- Students often include objectives in Chapter 1 but fail to explain how those objectives will be achieved in Chapter 3.

Helpful Academic Phrases

Research Design

- This study adopts...
- The research employs...
- An experimental approach was utilized...

Data Collection

- Data were collected from...
- The dataset was obtained from...

Methodology

- The proposed methodology consists of...
- The research process involves...

Evaluation

- Performance was evaluated using...
- The proposed model was assessed based on...

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Major Sections of Chapter 3

Evaluation Metrics



Why Evaluation Metrics Matter?

They determine whether the proposed solution **successfully achieved the research objectives**.

Research Design	Typical Evaluation Metrics
Experimental Study	Accuracy, Precision, Recall, F1-Score, RMSE, MAE
System Development	System Usability Scale, User Satisfaction, Response Time, Performance
Survey-Based Study	Mean, SD, Correlation
Case Study	Thematic Analysis, Expert Validation, Impact Measures

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Major Sections of Chapter 3

Chapter Summary



What to include in chapter summary section:

Briefly summarize how the chapter:

- ✓ **Explained** the selected research design and proposed methodology
- ✓ **Described** the data collection procedures
- ✓ **Identified** the tools, technologies, and computational resources
- ✓ **Defined** the testing methods and evaluation metrics
- ✓ **Provided** a transition to Chapter 4: Results and Discussion

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Do’s and Don’ts checklist for writing Chapter 3



Do’s (What to follow)	Don’ts (What to avoid)
Explain every step clearly	Skip methodology details
Use flowcharts	Use large text blocks
Justify chosen methods	Choose methods without explanation
Describe datasets fully	Hide dataset information
Link methods to objectives	Include unrelated procedures



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Purpose of Chapter 4



- **Present** and analyse findings clearly
- **Answer** research questions and objectives
- **Compare** findings with previous studies
- **Discuss** significant, unexpected, or contradictory results

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Key Questions Answered

- ✓ *What did the study find?*
- ✓ *What do the findings mean?*
- ✓ *How do the findings address the research objectives?*

Difference between Results, Analysis, and Discussion



Results: What was found?

Results present the actual evidence:

- Accuracy achieved by a model
- Survey response distributions
- System response time
- User-usability score
- Themes identified from interviews
- Screenshots of the developed system



Analysis: What patterns are visible?

Analysis examines:

- Differences between models or groups
- Highest and lowest values
- Trends and relationships
- Changes before and after implementation
- Recurring themes or patterns

Discussion: What do the results mean?

Discussion explains:

- Why a result may have occurred
- Whether the result answers the research question
- Whether an objective was achieved
- How the finding compares with earlier research
- What the result contributes to research or practice



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Major Sections of Chapter 4



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PLEASE NOTE

The **exact headings** may change based on the project type.
A system-development thesis may include implementation and interface results, while an experimental thesis may focus more heavily on model performance and comparative experiments.

Major Sections of Chapter 4

4.1 Overview

The opening should briefly explain:

- What this chapter **presents**
- Where the **results came from**
- How the **results** are **organised**
- Which **objectives/research questions** are **addressed**

Example



"This chapter presents and discusses the findings obtained from the implementation and evaluation of the proposed system. The results are organised according to the research objectives and include system implementation, performance evaluation, usability testing, and comparison with existing approaches."



Avoid

- Repeating the entire methodology
- Introducing new objectives
- Presenting conclusions before showing results
- Providing lengthy background information

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Major Sections of Chapter 4

4.2 Implementation or Experimental Setup

This section **explains what was implemented** or **executed** before the results were produced.

Include where applicable:

- Final model, system, or study configuration
- Implementation environment
- Software and hardware used
- Dataset partition or study groups
- Parameter settings
- System modules
- Testing environment
- Changes made during implementation

Experimental Study Setup (Experimental, AI, or Machine-Learning Studies)

Present:

- ✓ Dataset allocation for training, validation, and testing
- ✓ Final algorithms or models evaluated
- ✓ Parameter or hyperparameter settings
- ✓ Number or structure of experiments
- ✓ Baseline or benchmark methods
- ✓ Hardware and software environment
- ✓ Evaluation measures used



Example

"The experiments were implemented using Python and the selected machine-learning libraries. The dataset was divided into training and testing sets. The proposed model and baseline models were evaluated under the same experimental conditions using the metrics defined in Chapter 3."

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Major Sections of Chapter 4

4.3 Results and Discussion

Organize this section according to the research objectives or research questions.



For Each Objective:

1. Present the Evidence	Show relevant tables, graphs, results, screenshots, or statistical outputs.
2. Highlight main Result	Identify the key value, trend, difference, or finding.
3. Interpret the Finding	Explain what the result means, why it occurred, and its significance.
4. Link to the Objective	State how the finding answers the research question and whether the objective was achieved.

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Major Sections of Chapter 4

4.3 Results and Discussion

How to Discuss the Results?

Move Beyond Describing the Results

- The section should **not only state numbers** or repeat information from tables.
- The discussion should explain the **meaning and importance** of the findings.



A Strong Discussion Should Answer:

✓ What is the most important finding?
✓ What pattern, difference, or relationship is visible?
✓ Why might this result have occurred?
✓ Does the result support the expected outcome?
✓ Does it agree or disagree with previous studies?
✓ What does it contribute to the research problem?
✓ Was the research objective achieved fully or partially?
✓ Were there any unexpected or negative findings?

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Major Sections of Chapter 4

4.3 Results and Discussion

What Counts as Results and Discussion?

1. Experimental Study	Report model performance, compare approaches, and explain key results, errors, and unexpected findings.
2. System-Development Study	Present the system, evaluate its functions and performance, and discuss requirements, feedback, and limitations.
3. Interpret the Finding	Present key response patterns, statistical findings, and their implications for the research questions.

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Major Sections of Chapter 4

4.7 Comparison with Existing Studies



Compare your findings with **previous research**:

- Similarities and differences
- Why your results may differ
- Whether your findings support or contradict previous studies
- Contribution of your findings to existing knowledge

Sample Table for ML studies

Model	Accuracy	Precision	Recall	F1-Score
Baseline Model	Result	Result	Result	Result
Model A	Result	Result	Result	Result
Proposed Model	Result	Result	Result	Result

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Major Sections of Chapter 4

4.8 Chapter Summary





- **Presented the findings** obtained from the study
- **Addressed each research objective** through relevant evidence and results
- **Analysed and interpreted** the key findings
- **Discussed the significance** and implications of the results
- **Compared the findings** with existing studies and related literature
- **Established the foundation** for the conclusions, contributions, and future work presented in Chapter 5

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Do's and Don'ts checklist for writing Chapter 4

Do's (What to follow)	Don'ts (What to avoid)
Organise results by objectives	Present unrelated outputs
Use appropriate tables and figures	Insert screenshots without explanation
State and interpret important findings	Repeat every value in the table
Compare findings with previous studies	Claim superiority without fair comparison
Report unexpected results honestly	Hide weak or contradictory findings
Use the metrics defined in Chapter 3	Introduce unexplained new metrics
Distinguish evidence from interpretation	Mix personal opinion with findings





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Purpose of Chapter 5

- **Summarize** the overall research and key findings.
- **Explain** how the research objectives were achieved.
- **Highlight** the study's contributions and significance.
- **Discuss** limitations of the study.
- **Provide** recommendations and directions for future research.



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Key Questions Answered

- ✓ ***What did the study achieve?***
- ✓ ***What is the contribution of the study?***
- ✓ ***What limitations remain?***
- ✓ ***What should be done next?***

Major Sections of Chapter 5

5.1 Overview

5.2 Summary of the Study

5.3 Achievement of Project/Study Objectives

5.4 Project/Study Contributions

5.5 Project/Study Limitations

5.6 Recommendations and Future Work

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Major Sections of Chapter 5

5.1 Overview

The Overview should briefly state:

- The purpose of Chapter 5
- The main sections covered
- The connection with the findings in Chapter 4
- The chapter's role in concluding the research

Example

This chapter concludes the study by summarizing the research, reviewing the achievement of the research objectives, and highlighting the main contributions. The limitations of the study are also acknowledged, followed by recommendations and directions for future research.



Avoid

- Repeating the full research background
- Introducing new objectives
- Presenting new results
- Beginning an entirely new discussion
- Adding detailed tables or figures

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Major Sections of Chapter 5

5.2 Summary of the Study

Provide a **concise overview** of the complete research journey.

1. **Research Problem**
   What issue motivated the study?
2. **Research Aim and Objectives**
   What did the study intend to achieve?
3. **Research Approach**
   How was the problem addressed?
4. **Main Findings**
   What were the most important overall outcomes?
5. **Overall Significance**
   Why are these findings important?

Avoid Rewriting the Entire Thesis

NO

Do Not Repeat:

- Full research background from Chapter 1
- Detailed literature review from Chapter 2
- Step-by-step methodology from Chapter 3
- Complete results and tables from Chapter 4
- Long comparisons with every previous study
- Detailed technical explanations already presented

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Major Sections of Chapter 5

5.3 Achievement of Research Objectives

FOR EACH OBJECTIVE, EXPLAIN:

What was the objective?	Briefly restate the objective.
What was done?	Mention the corresponding research activity without repeating the methodology.
What was found?	State the key evidence or outcome.
Was the objective achieved?	Indicate whether it was fully achieved, partially achieved, or not achieved.
What does the achievement mean?	Explain its importance to the overall study.

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Major Sections of Chapter 5

5.4 Research Contributions

A **contribution** explains what the study adds beyond what was previously available.

Knowledge: New findings, relationships, or understanding.

Methodological: New or improved research methods, frameworks, or techniques.

Technical: New or improved models, algorithms, systems, or datasets.

Practical: Benefits or solutions for users, organisations, industry, or decision-makers.

Contribution Is Not the Same as Result

Key Formula

Finding + New Value + Beneficiary = Contribution



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Major Sections of Chapter 5

5.5 Research Limitations

- ✓ Identify conditions that may affect the **scope, findings, or generalisability** of the study.
- ✓ Common examples: **limited data/participants, single location, short study duration, limited resources, or restricted evaluation**.
- ✓ Explain how each limitation **may affect the study** and how future research can address it.

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Major Sections of Chapter 5

5.6 Recommendations and Future Work

- **Recommendations:** Suggest practical actions for stakeholders based on the research findings.
- **Future Work:** Suggest how future research can extend, improve, validate, or apply the study.
- Future work may address **limitations, unanswered questions, unexpected findings, technical improvements, or new contexts**.
- Keep suggestions **specific, realistic, and supported by the study**.

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Do's and Don'ts checklist for writing Chapter 5

Do's (What to follow)	Don'ts (What to avoid)
Summarise the overall study concisely	Repeat entire chapters
Conclude each research objective	Ignore objectives that produced weak results
Support conclusions with Chapter 4 evidence	Make unsupported claims
Clearly state contributions	Confuse findings with contributions
Acknowledge limitations honestly	Hide or exaggerate limitations
Link future work to limitations	Provide a random wish list
Make recommendations specific	Use vague statements





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Ensure appropriate citation

- ✓ ***Always cite works of other authors*** whose ideas, research findings, theories, or arguments or ***even Figures*** you have included or discussed in your literature review. ***Citing your resources*** is required for all kinds of writing projects.

- ✓ It is ***important to cite, not only direct quotes***, but ***paraphrased sentences*** as well. A paraphrased sentence is when a you take a passage from a source and rewrite it in your own words.

Example of quote: ***Evolution is a gradual process that "can act only by very short and slow steps" (Darwin, 1859).***

- ✓ ***Maintain only one citation style*** (APA, MLA, Chicago, IEEE) throughout your literature review.





https://www.sjsu.edu/writingcenter/

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APPENDICES



- ✓ Appendices Lengthy calculations, figures, raw data, computer programs/source codes, outputs, etc. are to be enclosed as appendices.
- ✓ They should **be titled and numbered** in chronological order and capital letters.
- ✓ The **appendices and their titles need to be listed in the Table of Contents**.
- ✓ Provide title for each appendix, for example "**Appendix 1**. **Questionnaire Sample**".

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Final Alignment Check - Trace the Complete Thesis Journey

Chapter 1
What did you intend to achieve?
Research problem, questions, and objectives

Chapter 3
How did you attempt to achieve it?
Research design, methods, tools, and evaluation

Chapter 4
What evidence was obtained?
Results, analysis, discussion, and comparison

Chapter 5
What was finally achieved?
Conclusions, contributions, limitations, and future work



The problem, objectives, methods, results, and conclusions must tell one consistent research story.

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Helpful Phrases for Writing Thesis

Helpful Phrases for Writing Background and others work

1. Previous literature

The literature review shows that __
Previous research showed __
Seminal contributions have been made by __
A series of recent studies has indicated that __
Several theories have been proposed to __, some focusing on __, others on __
There has been numerous studies to investigate __
This has been used in several studies to assess __
Previous studies have shown __
Several studies suggest that __
This has also been explored in prior studies by __
Prior research suggests that __
Previous studies have emphasized __
The majority of prior research has applied __
Most early studies as well as current work focus on __
For instance, the following studies were conducted on __
Studies of __are well documented, it is also well acknowledged that __
A number of authors have recognized __

2. Limitations of previous research

A number of questions regarding __ remain to be addressed.
A closer look to the literature on __, however, reveals a number of gaps and shortcomings.
This question has previously never been addressed because__
Most studies have relied on __
Previous studies by __ cannot be considered as conclusive because __
Previous studies have almost exclusively focused on __
This has been previously assessed only to a very limited extent because __
In the present studies __ were constrained to __
In previous studies were limited to __
Although results appear consistent with prior research, they appear inconsistent with __
These are previously unstudied because __
As far as we know, no previous research has investigated __
Moreover, although research has illuminated __ no study to date has examined __
Despite decades of research, this continues to be debated among __
This section points out some of the problems encountered in the extant research.
Although there are many studies, the research in __ remains limited.

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https://www.ref-n-write.com/blog/literature-review-section-academic-phrases/

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Helpful Phrases for Writing Thesis

Helpful Phrases for Writing Background and others work

3. Research questions

More specific research questions will be introduced and investigated in __
A further question is whether __
Finally, another promising line of research would be __
The study addresses several further questions on __
Some of the interesting questions in this context are __
In order to address the questions outlined above, we report here __
These questions are of central interest as much recent research in __
Furthermore, __ is arguably an important question to be addressed
The question now is how __ can be used to explain __
Study addresses the research question __
In order to properly address this question, we __
An important question associated with __ is __
A critical open question is whether __
A still unsolved question is whether __
This remains an open question as __
This question has previously never been addressed because __
This study offers a test of __ research question
Study addresses the research question __

4. Research to be explored

A more systematic and theoretical analysis is required for __
As the authors note earlier, more work is necessary to__
Additional studies to understand more completely the key tenets of __ are required.
The unexpected findings signal the need for additional studies to understand more about __
This paper addresses __, so far lacking in the scientific literature.
A new approach is therefore needed for __
One of the tough challenges for all researchers in this domain is __

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https://www.ref-n-write.com/blog/literature-review-section-academic-phrases/

Helpful Phrases for Writing Thesis

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Helpful Phrases for Writing Chapter 4

Presenting Results

- The results indicate that...
- Figure 4.X illustrates...
- Table 4.X presents...
- The highest performance was achieved by...
- A noticeable difference was observed between...

Analysing Results

- The findings reveal a consistent pattern...
- The result demonstrates that...
- This improvement may be attributed to...
- A possible explanation for this result is...
- The observed difference suggests that...

Connecting to Objectives

- This finding addresses Research Objective 1 by...
- The result provides an answer to Research Question 2...
- These findings demonstrate that the proposed solution...

Comparing with Literature

- This finding is consistent with...
- The result supports the observations reported by...
- In contrast to previous research...
- The current study achieved a different outcome because...
- This finding extends previous research by...

Reporting Limitations

- Although the result is promising...
- The performance was affected by...
- The finding should be interpreted with caution because...
- Further analysis is required to determine...

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Helpful Phrases for Writing Thesis

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Helpful Academic Phrases for Chapter 5

Summarising the Study
- This study investigated...
- The research aimed to...
- The study addressed the problem of...
- The research adopted...

Concluding the Findings
- The findings demonstrate that...
- The study concludes that...
- The results provide evidence that...
- The findings indicate the potential of...

Addressing Objectives
- Research Objective 1 was achieved through...
- The findings related to Objective 2 demonstrate...
- The evidence presented in Chapter 4 confirms that...
- This objective was partially achieved because...

Stating Contributions
- This study contributes to existing knowledge by...
- The main practical contribution of the study is...
- The proposed approach extends previous research by...
- The developed system provides...

Acknowledging Limitations
- This study is subject to several limitations...
- The findings should be interpreted within...
- One limitation of the study is...
- The generalisability of the findings may be affected by...

Suggesting Future Work
- Future research should investigate...
- Further studies could extend...
- The proposed approach could be improved by...
- Future work may evaluate...
- A wider evaluation is recommended

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Dissertation/Thesis Formatting Quick Rules



Figure captions → Below the figure

Table captions → Above the table

All tables/figures → Must be numbered, cited, and discussed in text

Placement → Insert close to first mention in text

Consistency → Same style/format for all captions

Summary rule → Chapter 1 ends with *thesis outline*; Chapter 5 ends with *recommendations.*

→ *Chapter 2, 3, 4 should end with Summary section.*

Length

The maximum length of the report, excluding appendices is as follows:

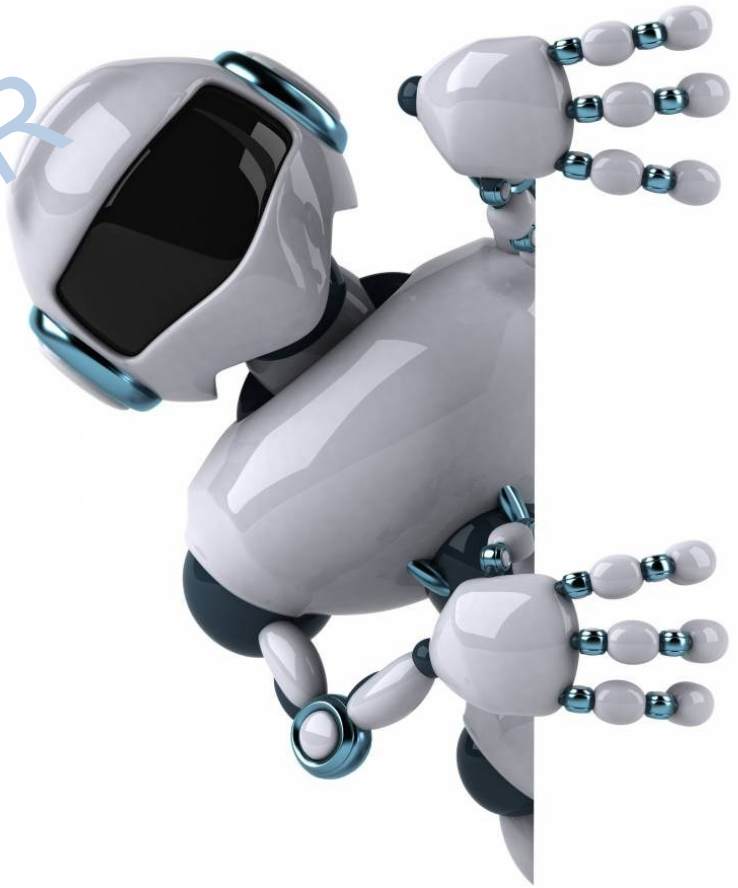
Interim Report	20 pages
Technical Paper	06 pages
Dissertation	50 pages

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Thank You!